



Development of the Sustainable Living Quality Index (SLQI)

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(software developement)

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Development of the Sustainable Living Quality Index (SLQI)

Introduction

This report presents the development of the Sustainable Living Quality Index (SLQI). The Purpose of the index is to compare countries using a group of indicators related to sustainable living quality. Instead of using only one variable, the SLQI combines several dimensions that describe different parts of living conditions and sustainability.

The index was developed using country-level datasets. The selected indicators were grounded into five main dimensions: economic conditions, health and education, infrastructure, governance, and environment. These dimensions were chosen because they provide a broad view of living quality. For example, economic indicators describe material conditions, while health and education indicators describe human development outcomes. Infrastructure indicators show access to basic services, governance indicators show the quality and stability of institutions, and environmental indicators show whether countries are performing well in relation to environmental sustainability.

This follows the full development process of the SLQI. It begins with

- > data selection and cleaning.
- > then explains missing data treatment, outlier checking,
- > multivariate analysis,
- > normalization,
- > weighting and aggregation,
- > link with the Human Development Index (HDI),
- > and visualization.

The final output is a country-level SLQI score and rank.

Theoretical Framework

Concept of the Sustainable Living Quality Index

The Sustainable Living Quality Index is based on the idea that living quality can't be measured by one indicator alone. A country may have a high-income level but still have environmental problems, weak governance, or unequal access to basic services. In the

same way, a country may perform well in environmental indicators but still have lower performance in income, infrastructure, or health outcomes.

For this reason, the SLQI uses a multidimensional structure. It combines indicators into one composite index:

- Economic
- Health and education (Social)
- Infrastructure
- Environmental
- Governance

Each indicators measures a different part of sustainable living quality. By combining them, the index provides a more comprehensive reflection than a single indicator such as GDP per capita.

Index Dimensions and Indicators

The selected indicators are shown in table. These indicators were taken from the datasets used in the notebooks. Each indicator was assigned to one of the five dimensions.

Dimension	Indicators Used	Directions
Economic	GDP per capita, inflation, unemployment rate	GDP per capita is positive; inflation and unemployment are negative
Health and Education	Life expectancy, secondary school enrolment	Positive
Infrastructure	Electricity access, drinking water access, sanitation access	Positive
Governance	Government effectiveness, political stability, rule of law	Positive
Environment	Forest area, PM2.5 exposure, renewable energy	Forest area and renewable energy are positive; PM2.5 exposure is negative

Positive indicators mean that higher values represent better performance.

Negative indicators means that lower values represent better performance.

For example: a higher life expectancy is a positive outcome, but higher PM2.5 exposure is a negative outcome because it represents higher air pollution.

Methodology

Overview of the Index Development Process

The SLQI was developed through several stages. First started with loading and selecting relevant datasets. The selected variables were then cleaned, combined, and checked for missing values. After that, the indicators were normalized so they could be compared on the same scale. Dimension scores were calculated, and the final SLQI score was produced using equal weighting.

The full workflow was:

Data loading -> Indicator selection -> Dataset merging -> Missing data treatment -> Outlier checking -> Multivariate analysis -> Normalization -> Weighting and aggregation -> Link to other indices -> Visualization

Five datasets were loaded at the start of the project. These datasets covered economic, health and education, infrastructure, environment, and governance indicators. The missing value symbol “..” was treated as a missing value when the datasets were loaded.

Country Name	Country Code	Series Name	Series Code	2020 [YR2020]	2021 [YR2021]	2022 [YR2022]	2023 [YR2023]	2024 [YR2024]	2025 [YR2025]
Afghanistan	AFG	GDP per capi	NY.GDP.PCA	510.787063	356.496214	357.261153	413.757895	..	..
Afghanistan	AFG	Inflation, con	FP.CPI.TOTL	5.60188791	5.13320341	13.7121024	-4.6447087	-6.6011856	..
Afghanistan	AFG	Unemploye	SL.UEM.TOTL	11.71	12.006	14.1	14.008	13.687	13.351
Albania	ALB	GDP per capi	NY.GDP.PCA	6027.91351	7242.45513	7756.96189	9730.86922	11377.7757	..
Albania	ALB	Inflation, con	FP.CPI.TOTL	1.62088662	2.04147163	6.72520272	4.75834555	2.21587375	..
Albania	ALB	Unemploye	SL.UEM.TOTL	11.639	11.474	10.785	10.669	10.689	10.926
Algeria	DZA	GDP per capi	NY.GDP.PCA	3743.54195	4160.55927	4960.30334	5370.47724	5752.99077	..
Algeria	DZA	Inflation, con	FP.CPI.TOTL	2.41513094	7.22606307	9.26551552	9.32217376	4.04611455	..
Algeria	DZA	Unemploye	SL.UEM.TOTL	14.08	13.656	12.382	11.645	11.655	11.633
American Sai	ASM	GDP per capi	NY.GDP.PCA	14489.2587	15236.1605	18017.4589	..	..	..
American Sai	ASM	Inflation, con	FP.CPI.TOTL	..	..	..	..	..	..
American Sai	ASM	Unemploye	SL.UEM.TOTL	..	..	..	..	..	..
Andorra	AND	GDP per capi	NY.GDP.PCA	37361.107	42425.7066	42414.048	46812.4261	49303.6492	..
Andorra	AND	Inflation, con	FP.CPI.TOTL	..	..	..	..	..	..
Andorra	AND	Unemploye	SL.UEM.TOTL	..	..	..	..	..	..
Angola	AGO	GDP per capi	NY.GDP.PCA	1759.3562	2303.90813	3682.11315	2916.13663	2665.87445	..
Angola	AGO	Inflation, con	FP.CPI.TOTL	22.2715392	25.7542949	21.3552901	13.6441018	28.2404949	..
Angola	AGO	Unemploye	SL.UEM.TOTL	16.696	15.799	14.124	14.051	14.02	14.108
Antigua and B	ATG	GDP per capi	NY.GDP.PCA	15369.6082	17348.6007	20105.1989	21494.5474	23542.4527	..

Figure 1 e.g., missing value symbol .. was treated as a missing value

Use of Recent Available Data

The data mainly used the most recent available data. In many cases, 2024 was used first. If 2024 data was missing, earlier available values were used as a backfill. This approach helped keep the index as recent as possible while reducing missing data.

- For the economic indicators, the value from 2024 was used first. If it was missing, values from 2023, 2022, 2021, or 2020 were used.
- For life expectancy, 2024 was used first and 2023 was used as a backfill.
- For secondary enrolment, the latest available value from 2024 to 2020 was used.
- For infrastructure and governance indicators, 2024 was used first and 2023 was used as a backfill.
- For environmental indicators, different years were used depending on data availability. Forest area used 2023 first and 2022 as a backfill. PM2.5 exposure used 2020 because that was the year with available data for most countries. Renewable energy used 2022 first and 2021 as a backfill.

Data Selection and Cleaning

Selected Variables

The selected variables were grouped into the five SLQI dimensions. Total variable was 14.

- The economic dimension included GDP per capita, inflation, and unemployment rate.
- The health and education dimension included life expectancy and secondary school enrolment.
- The infrastructure dimension included electricity access, drinking water access, and sanitation access.
- The governance dimension included government effectiveness, political stability, and rule of law.
- The environment dimension included forest area, PM2.5 exposure, and renewable energy.

These variables were selected because they provide a balanced view of country-level living quality and sustainability. The selection also considered data availability. For example, tertiary enrolment was removed because it had too many missing values even after using the latest available data from 2020 to 2024.

Combining the Datasets

After selecting and cleaning the relevant variables from each dataset, the five datasets were merged into one combined dataset. Country and Country Code were used as the

common keys. An outer merge was used so that countries from all datasets could be included before handling miss data.

The combined dataset contained: 225 countries and 14 variables

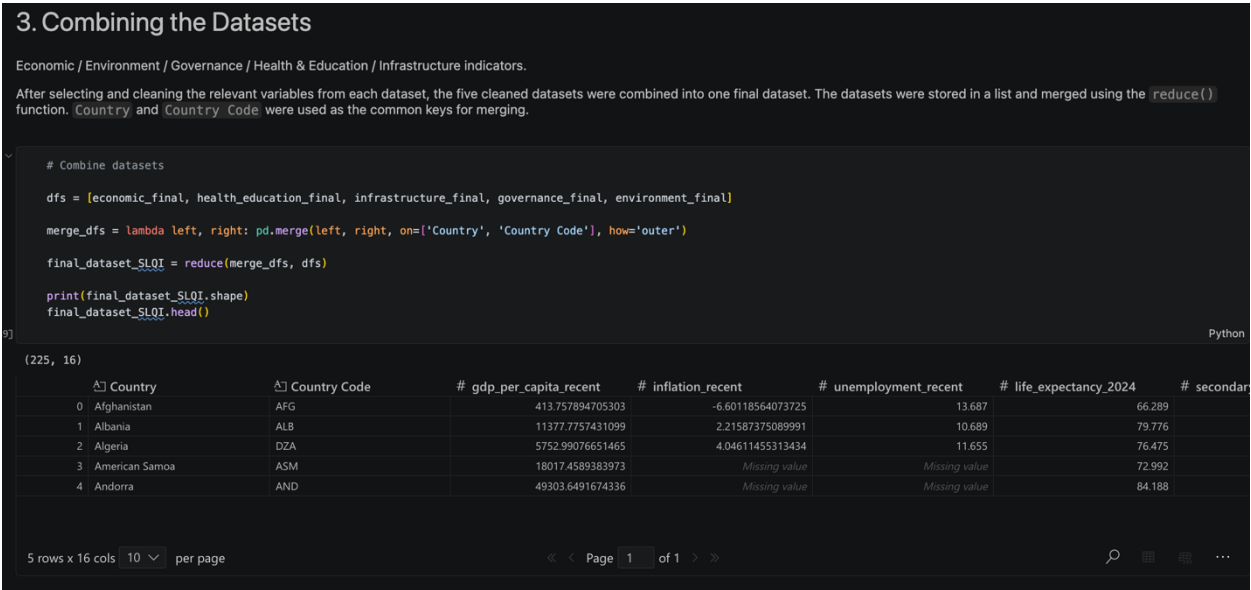


Figure 2 Combining the datasets

Renaming Variables

The variable names were renamed into shorter and more consistent names. For example, GDP per capita (current US\$) was renamed as `gdp_per_capita`, and so on. That would made the dataset easier to read and easier to use in later stages. I removed year labels from the final column names because some indicators used the latest available value rather than a single fixed year.

Missing Data Before Cleaning

After merging the datasets, I checked missing values for each variable. Some variables had more missing values than others.



Figure 3 Number of Missing Value

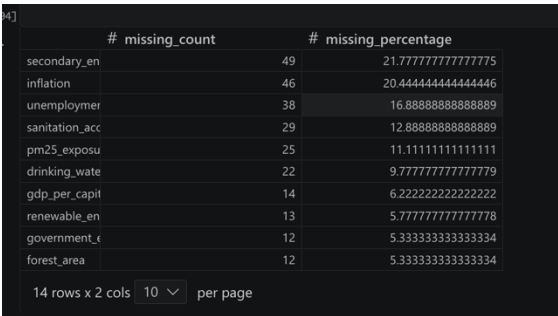


Figure 4 Percentage of each missing value

The number and percentage of missing values were calculated for each variable. It helps to identify which indicators have the most missing data and may need further cleaning or removal. Because Tertiary enrolment was contained a high number of missing values even after using the latest available data from 2020 to 2024. (70 missing values and over 30%) so that's why I removed this.

And then countries with more than four missing indicators were removed. This was used because these countries did not have enough data to calculate a reliable index score.

After cleaning from 225 countries to 206 countries.



The screenshot shows a Jupyter Notebook interface. At the top, there is a table with 4 columns: Country, Country Code, # missing_values_count, and # available_values_count. The table lists 10 countries, including Jersey, Martinique, Cook Islands, Taiwan, China, Reunion, Niue, French Guiana, Anguilla, Channel Islands, and Sint Maarten (Dutch part). Below the table, there is a pagination bar indicating 100 rows x 4 cols, 10 per page, and Page 1 of 10. Below the pagination bar, there is a bullet point stating: "I also checked missing values at the country level. These countries may need to be removed later if there is not enough data to calculate a reliable index score." Below this, there is a code cell containing Python code to remove countries with too many missing values. The code sets max_missing_allowed = 4, filters the final_dataset_SLOI to keep only countries with missing_values_count <= 4, and prints the original and cleaned country counts. The output of the code cell shows: Original countries: 225, Countries after cleaning: 206.

	Country	Country Code	# missing_values_count	# available_values_count
102	Jersey	JEY	11	3
129	Martinique	MTQ	11	3
47	Cook Islands	COK	11	3
197	Taiwan, China	TWN	11	3
166	Reunion	REU	11	3
150	Niue	NIU	11	3
71	French Guiana	GUF	11	3
6	Anguilla	AIA	11	3
40	Channel Islands	CHI	10	4
179	Sint Maarten (Dutch part)	SXM	9	5

```

# Remove countries with too many missing values
max_missing_allowed = 4
final_dataset_clean = final_dataset_SLOI[final_dataset_SLOI['missing_values_count'] <= max_missing_allowed].copy()

print(f"Original countries: {len(final_dataset_SLOI)}")
print(f"Countries after cleaning: {len(final_dataset_clean)}")

final_dataset_clean.head(206)

```

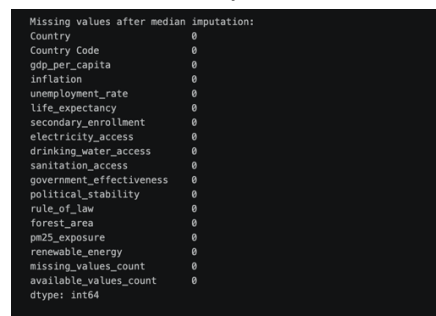
Original countries: 225
Countries after cleaning: 206

Figure 3 Removed more than 4 missing indicators

Median Imputation

After removing countries with too many missing indicators, some missing values still remained. These missing values were filled using median imputation. Median imputation was used because it is simple, clear, and less affected by extreme values than the mean.

After median imputation, the final cleaned dataset had no missing values.



The screenshot shows a Jupyter Notebook output. It displays a list of variables and their corresponding missing values after median imputation. All values are 0, indicating that no missing values remain. The variables listed are: Country, Country Code, gdp_per_capita, inflation, unemployment_rate, life_expectancy, secondary_enrollment, electricity_access, drinking_water_access, sanitation_access, government_effectiveness, political_stability, rule_of_law, forest_area, pm25_exposure, renewable_energy, missing_values_count, available_values_count, and dtype: int64.

```

Missing values after median imputation:
Country          0
Country Code     0
gdp_per_capita   0
inflation        0
unemployment_rate 0
life_expectancy  0
secondary_enrollment 0
electricity_access 0
drinking_water_access 0
sanitation_access 0
government_effectiveness 0
political_stability 0
rule_of_law      0
forest_area      0
pm25_exposure    0
renewable_energy 0
missing_values_count 0
available_values_count 0
dtype: int64

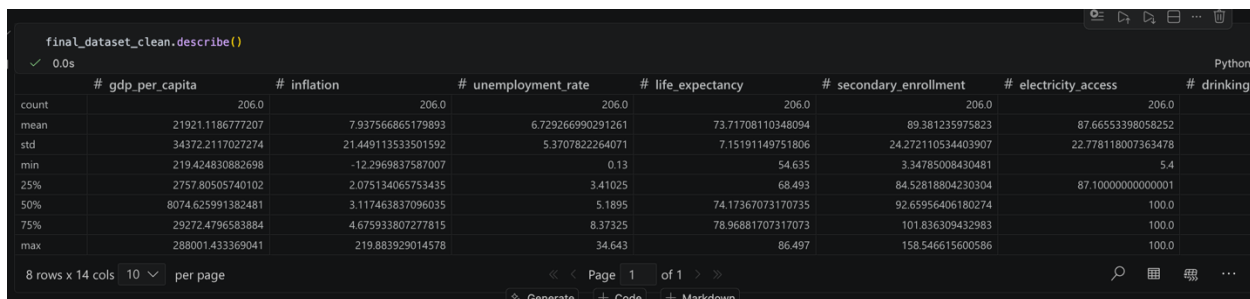
```

Figure 4 Use Median Imputation to fill missing values

Checking for Outliers

After handle missing data, I checked for outliers and invalid values. The main aim of this step was not to remove all unusual values. Instead, it was used to identify values that may need review.

First, summary statistics were checked using `describe()`. This showed the minimum, maximum, mean, and quartile values for each variable. Then, variables that should not contain negative values were checked. These included GDP per capita, unemployment rate, life expectancy, secondary enrolment, electricity access, drinking water access, sanitation access, forest area, PM2.5 exposure, and renewable energy.

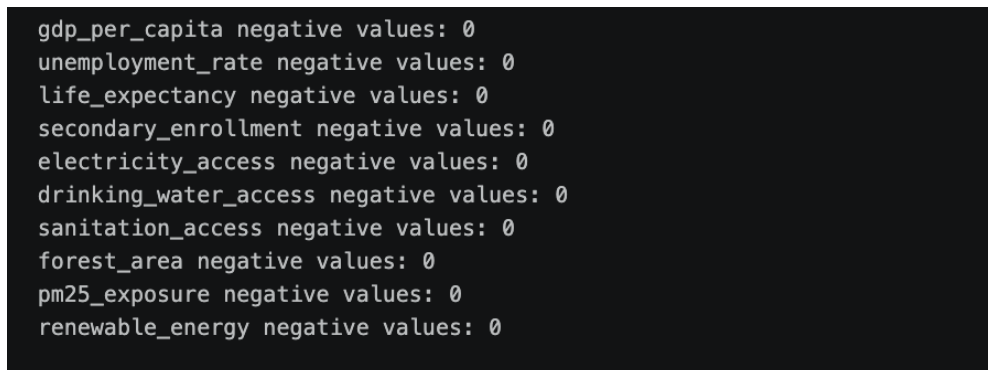


The screenshot shows a Jupyter Notebook interface with a code cell containing `final_dataset_clean.describe()`. The output is a summary statistics table for 14 variables. The variables are: `# gdp_per_capita`, `# inflation`, `# unemployment_rate`, `# life_expectancy`, `# secondary_enrollment`, `# electricity_access`, `# drinking_water_access`, `# sanitation_access`, `# forest_area`, `# pm25_exposure`, `# renewable_energy`, and three unnamed variables. The table includes rows for count, mean, std, min, 25%, 50%, 75%, and max. The interface also shows a progress bar at the top, a search bar, and a table view icon.

	# gdp_per_capita	# inflation	# unemployment_rate	# life_expectancy	# secondary_enrollment	# electricity_access	# drinking_water_access	# sanitation_access	# forest_area	# pm25_exposure	# renewable_energy		
count	206.0	206.0	206.0	206.0	206.0	206.0	206.0	206.0	206.0	206.0	206.0		
mean	21921.1186772207	7.937566865179893	6.729266990291261	73.71708110348094	89.381235975823	87.66553398058252	87.66553398058252	87.66553398058252	87.66553398058252	87.66553398058252	87.66553398058252		
std	34372.2117027274	21.449113533501592	5.3707822264071	7.15191149751806	24.272110534403907	22.778118007363478	22.778118007363478	22.778118007363478	22.778118007363478	22.778118007363478	22.778118007363478		
min	219.424830882698	-12.2969837587007	0.13	54.635	3.34785008430481	5.4	5.4	5.4	5.4	5.4	5.4		
25%	2757.80505740102	2.075134065753435	3.41025	68.493	84.52818804230304	87.10000000000001	87.10000000000001	87.10000000000001	87.10000000000001	87.10000000000001	87.10000000000001		
50%	8074.625991382481	3.117463837096035	5.1895	74.17367073170735	92.65956406180274	100.0	100.0	100.0	100.0	100.0	100.0		
75%	29272.4796583884	4.675933807277815	8.37325	78.96881707317073	101.836309432983	100.0	100.0	100.0	100.0	100.0	100.0		
max	288001.433369041	219.883929014578	34.643	86.497	158.546615600586	100.0	100.0	100.0	100.0	100.0	100.0		

Figure 5 summary statistics describe

No invalid negative values were found in these.



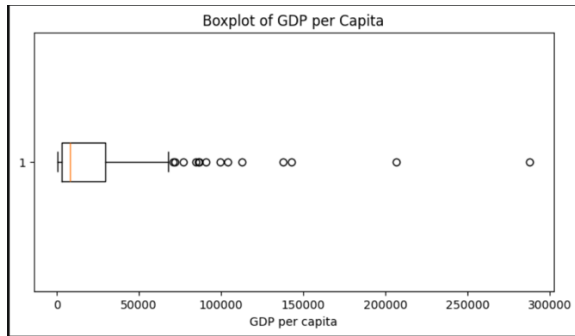
The screenshot shows a Jupyter Notebook interface with a code cell containing a loop that checks for negative values in various variables. The output is a list of variables followed by the text "negative values: 0".

```
gdp_per_capita negative values: 0
unemployment_rate negative values: 0
life_expectancy negative values: 0
secondary_enrollment negative values: 0
electricity_access negative values: 0
drinking_water_access negative values: 0
sanitation_access negative values: 0
forest_area negative values: 0
pm25_exposure negative values: 0
renewable_energy negative values: 0
```

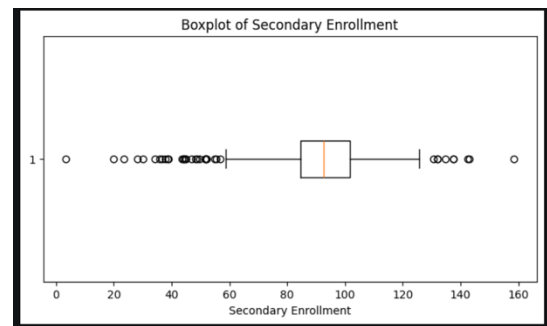
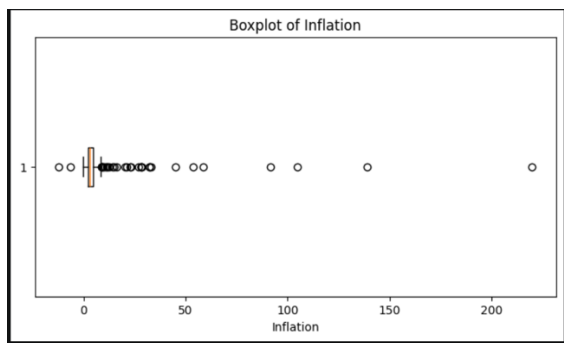
Figure 6 Non-Negative variables

Then used the IQR method to identify potential outliers. Some variables had more potential outliers than others.

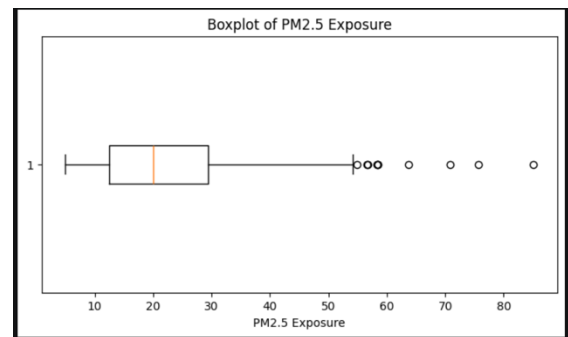
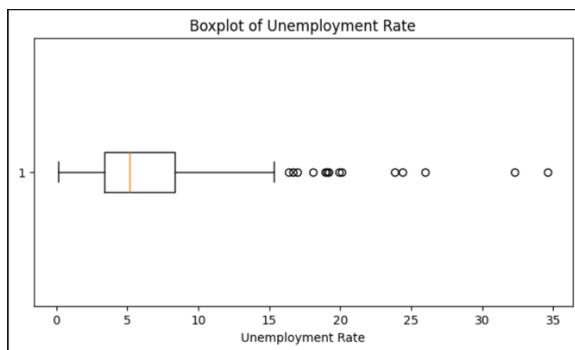
GDP Per Capita



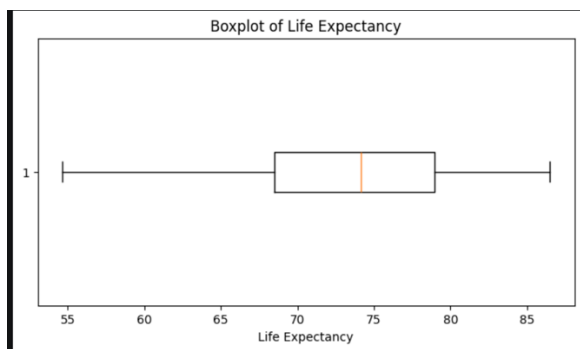
Inflation



Unemployment Rate



Life expectancy



Variable	# Outlier Count	# Lower Bound	# Upper Bound
5 electricity_access	39	67.75000000000003	119.35
4 secondary_enrollment	36	58.56600595628311	127.79849151900294
1 inflation	31	-1.826065546533135	8.577133419564385
6 drinking_water_access	22	70.81605066188715	117.1267169235692
7 sanitation_access	15	31.393137817078987	139.52079478755354
0 gdp_per_capita	14	-37014.20684408006	69044.49155986948
2 unemployment_rate	14	-4.034250000000002	15.817750000000002
12 pm25_exposure	10	-12.909103958721412	54.839929352359086
9 political_stability	1	-2.7216271625	2.8932695375
3 life_expectancy	0	52.7792743902439	94.68254268292682
8 government_effectiveness	0	-2.7572588124999995	2.7820676874999997
10 rule_of_law	0	-3.2456285874999997	3.0976997125
11 forest_area	0	-45.74768696073331	107.01012548829911
13 renewable_energy	0	-49.0375	105.26250000000002

Figure 7 Outliers

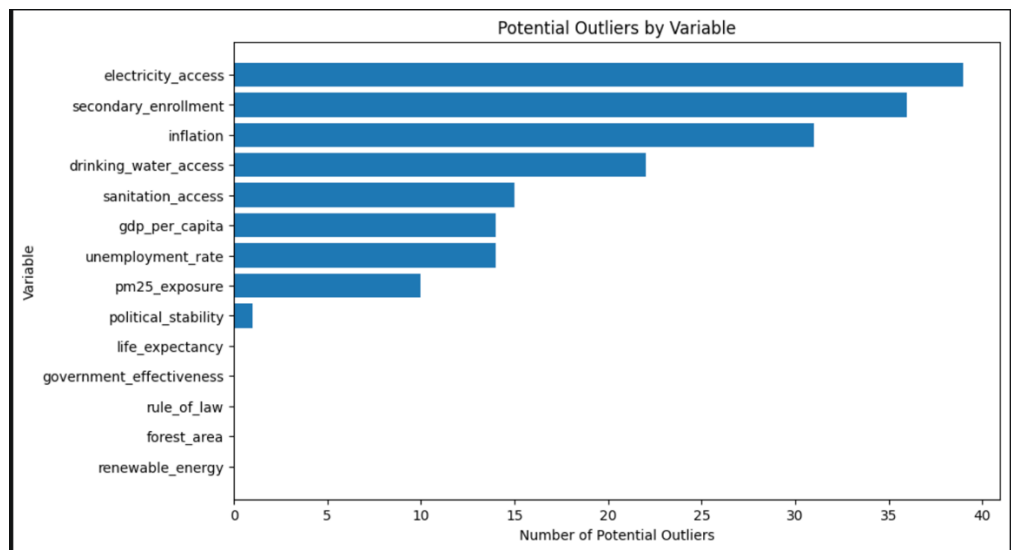


Figure 8 Potential Outliers by Variable

The outliers were not automatically removed. This is because some extreme values can represent real country-level differences. For example, some countries naturally have very high GDP per capita, high inflation, or low infrastructure access. If removing these values could reduce the accuracy of the index.

Multivariate Analysis

Descriptive Statistics and Distributions

I used histograms to examine the distribution of the selected indicators. The histograms showed that several variables were not normally distributed. GDP per capita and inflation were strongly right skewed. This means that most countries had relatively low or moderate

values, while a smaller number of countries had very high values. this is making sense because European countries are high GDP per capita.

Infrastructure indicators such as electricity access, drinking water access, and sanitation access were mostly concentrated at higher values. This suggests that many countries in the dataset had relatively strong access to basic services. Life expectancy was more evenly distributed, with most countries falling around the middle to upper range.

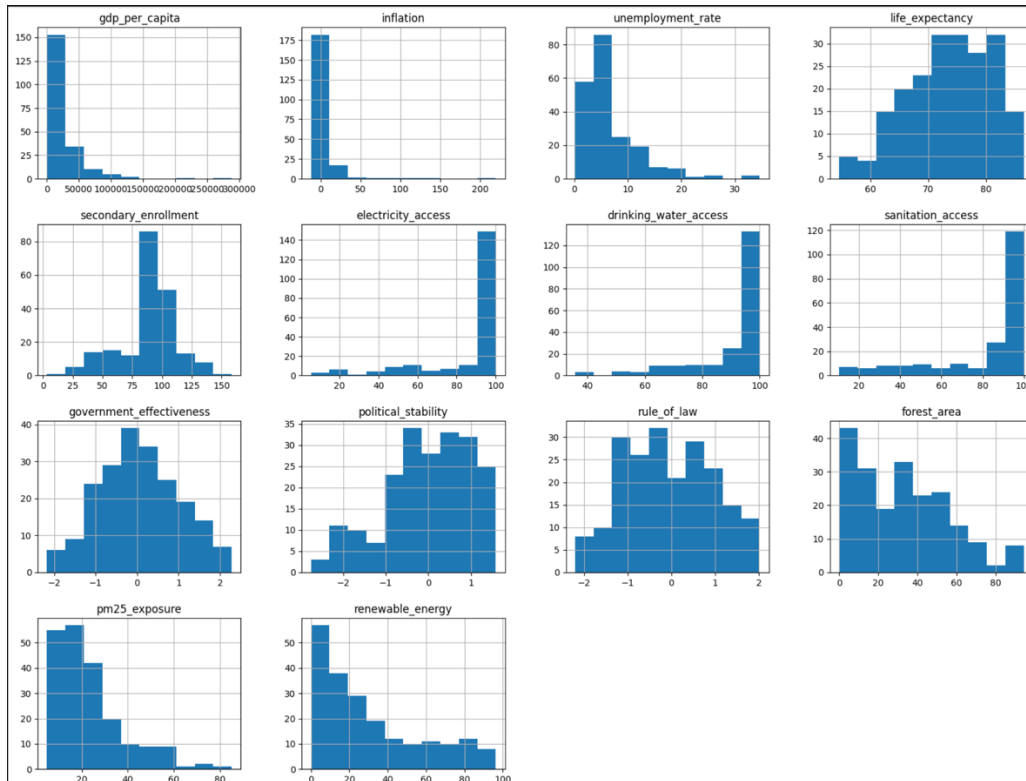


Figure 9 histograms

I was created correlation matrix to examine linear relationships between the indicators.

- A correlation value close to 1 means a strong positive relationship.
- A value close to -1 means a strong negative relationship.
- A value close to 0 means a weak linear relationship.

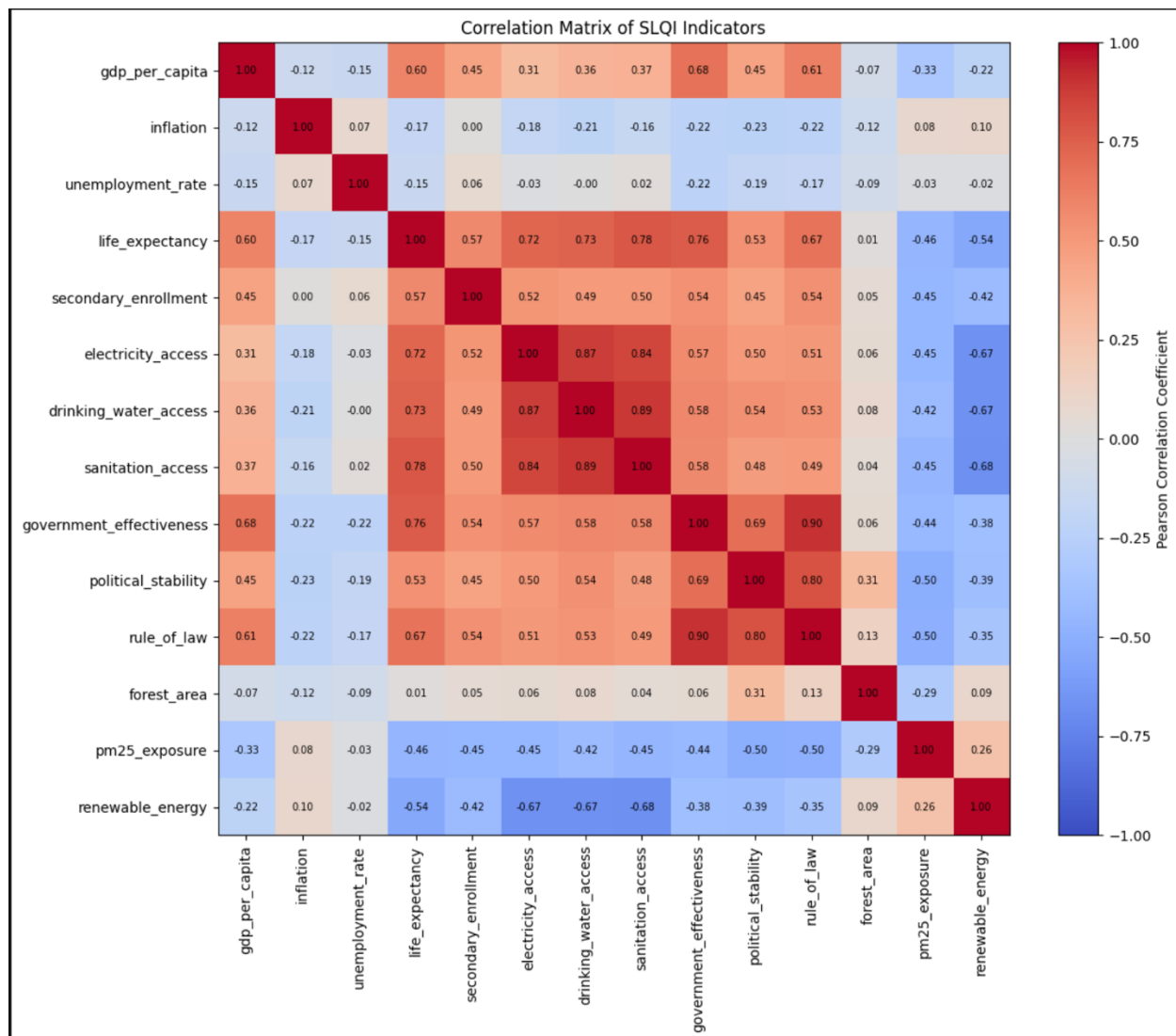


Figure 10 Correlation Matrix Heatmap

The strongest positive relationship was between government effectiveness and rule of law, with a correlation of 0.90. This means that countries with more effective governments also tended to have stronger rule of law.

Drinking water access and sanitation access also had a strong positive correlation of 0.89. Electricity access and drinking water access had a correlation of 0.87, while electricity access and sanitation access had a correlation of 0.84.

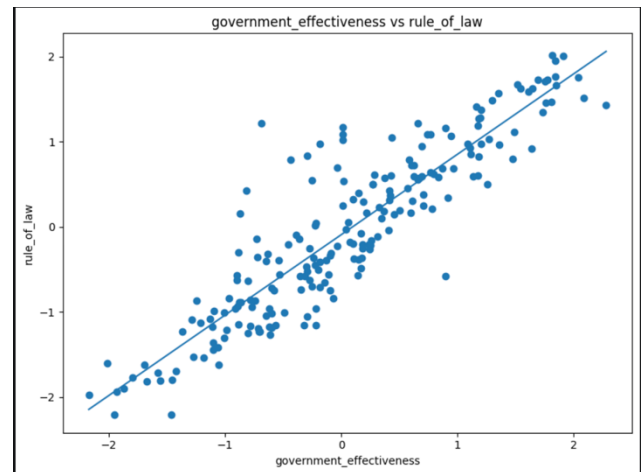
The strongest negative relationships involved renewable energy and some infrastructure or living quality indicators. For example, sanitation access and renewable energy had a correlation of -0.68. Drinking water access and renewable energy had a correlation of -0.67. This does not mean renewable energy reduces living quality. It may reflect that some

countries with high renewable energy shares rely more on traditional biomass or less industrialized energy systems.

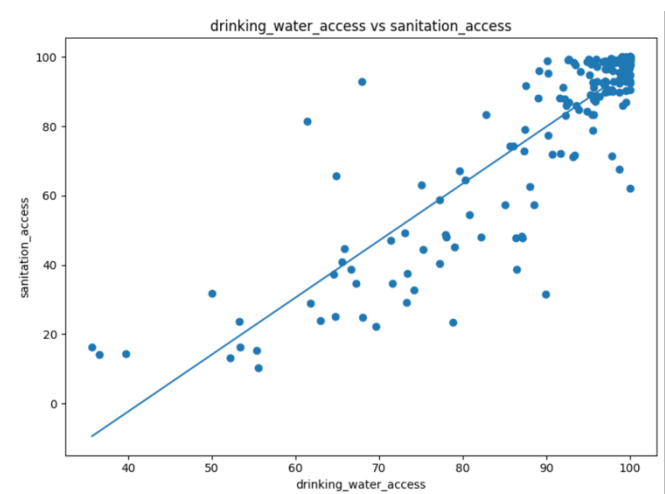
PM2.5 exposure also showed negative relationships with several living quality indicators. life expectancy and PM2.5 exposure had a negative relationship. This suggests that countries with higher pollution exposure tend to have lower life expectancy.

Scatter Plot Analysis

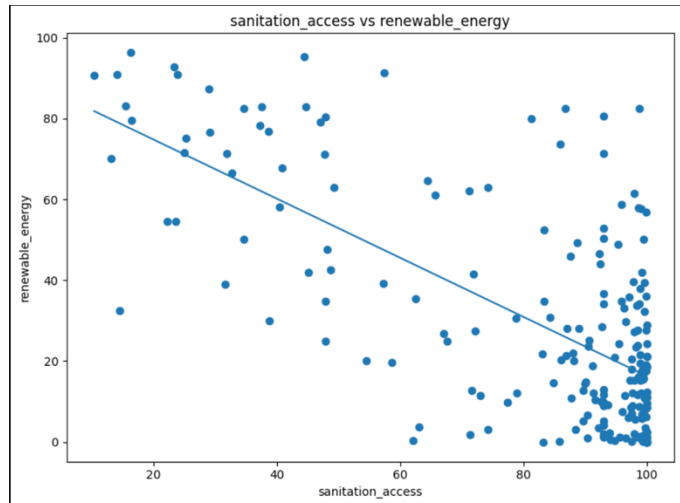
Government effectiveness and rule of law showed a strong positive relationship. This means that countries with more effective public institutions also have stronger legal systems.



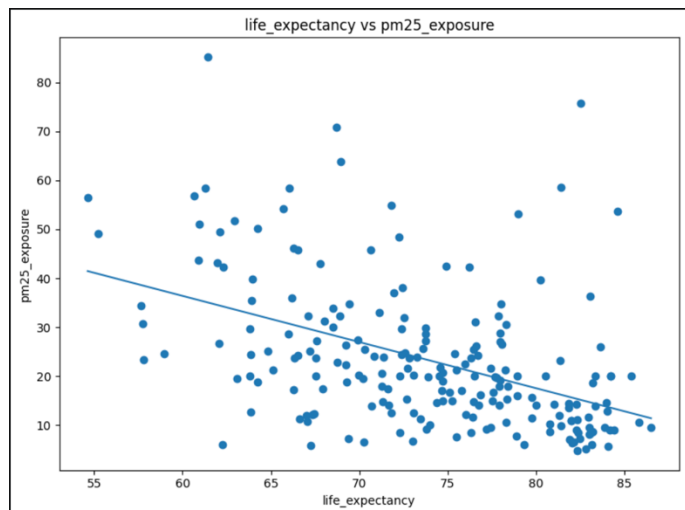
Drinking water access and sanitation access also showed a strong positive relationship. Countries with better access to clean drinking water usually also had better sanitation services.



Sanitation access and renewable energy showed a moderate negative relationship. This result needs careful interpretation. It does not show a direct negative effect of renewable energy. It may reflect differences between highly industrialized countries and countries that depend more heavily on renewable.



Life expectancy and PM2.5 exposure showed a negative relationship. This suggests that higher pollution exposure is generally associated with lower life expectancy.



Principal Component Analysis (PCA)

Before using PCA, I standardized the indicators. I did this because the indicators had different units and scales.

The result showed that PC1 explained 47.29% of the total variance. This means PC1 captured nearly half of the information in the dataset. PC2 explained 11.48%, PC3 explained 8.75%, PC4 explained 7.75%, and PC5 explained 6.30%.

	Principal Component	# Explained Variance Ratio	# Explained Variance Perce...
0	PC1	0.47285361616105986	47.28536161610599
1	PC2	0.11484330448952475	11.484330448952475
2	PC3	0.08745419493283085	8.745419493283086
3	PC4	0.0774799968821213	7.74799968821213
4	PC5	0.06303352449050306	6.303352449050307
5	PC6	0.03834699172047566	3.834699172047566
6	PC7	0.03469719428756364	3.469719428756364
7	PC8	0.03404315179014928	3.404315179014928
8	PC9	0.026302603362851035	2.6302603362851036
9	PC10	0.017711583608145585	1.7711583608145585

14 rows x 3 cols 10 per page << < Page 1 of 2 > >>

The first five PCA explained about 81.57% of the total variance together. From this result, I through that five components were enough to keep most of the important information from the original indicators.

I also checked the loading scores to understand which indicators had more influence on PC1. PC1 was linked with life expectancy, government effectiveness, drinking water access, sanitation access, electricity access, rule of law, political stability, secondary enrolment, renewable energy, and GDP per capita.

...	# PC1
life_expectancy	0.34220781352068247
government_effectiveness	0.3308722769186929
drinking_water_access	0.32989348221201403
sanitation_access	0.3271771577728951
electricity_access	0.3231415531023713
rule_of_law	0.32005143499719246
political_stability	0.2956209422677627
secondary_enrollment	0.26596987189110566
renewable_energy	-0.25404199216743606
gdp_per_capita	0.24268913146987528

...	# PC2
unemployment_rate	0.3898656366019238
renewable_energy	-0.38512068129381566
forest_area	-0.3319657881360734
rule_of_law	-0.3126869889137621
sanitation_access	0.30165497328575636
political_stability	-0.28966176916760766
electricity_access	0.28097379710934184
drinking_water_access	0.26831323756729447
gdp_per_capita	-0.248451550221539
government_effectiveness	-0.2446271007914941

...	# PC3
forest_area	0.7115738103349449
gdp_per_capita	-0.4200315743424784
pm25_exposure	-0.3097206847463021
inflation	-0.2095151786235087
government_effectiveness	-0.20876873604241694
unemployment_rate	0.18448929772297396
political_stability	0.14838706216612896
life_expectancy	-0.14704718015346901
drinking_water_access	0.12668226258216203
electricity_access	0.12237048900708734

...	# PC4
inflation	0.6124533079460652
unemployment_rate	0.4791478402737574
secondary_enrollment	0.3918756552123955
pm25_exposure	-0.36551374649043256
renewable_energy	0.1549466366689117
drinking_water_access	-0.15406912828448197
gdp_per_capita	0.13973005074830955
electricity_access	-0.12789375551284998
sanitation_access	-0.09777539414833768
forest_area	0.08199378447938813

...	# PC5
inflation	0.6602854057299998
unemployment_rate	-0.6598443803985933
forest_area	0.19423578948637718
gdp_per_capita	-0.1756616463337404
renewable_energy	-0.13299818710492095
electricity_access	0.11305792607034752
rule_of_law	-0.10984882798186085
sanitation_access	0.07638665220238494
government_effectiveness	-0.06640894867748776
drinking_water_access	0.0600470910066335

Most of these indicators are related to development and living conditions. Because of this, PC1 as a general development and living quality component.

Cluster Analysis

I worked on K-Means cluster analysis to group countries with similar indicator patterns. I used the first five principal components for clustering because they explained more than 80% of the total variance.

I selected four clusters for this analysis. The silhouette score was 0.297. the score is not very high, so the clusters were not strongly separated. Because of this, I treated the cluster analysis as an exploratory method, rather than a final way to classify countries.

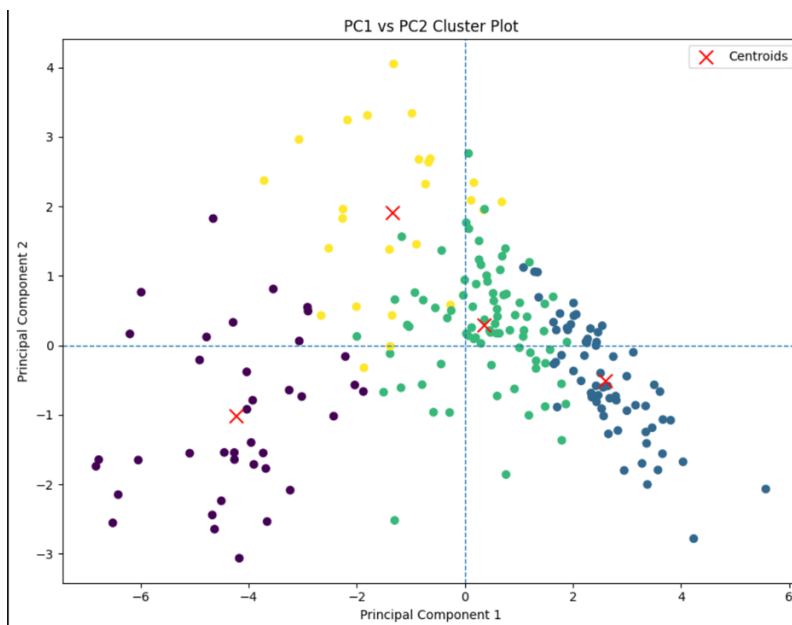


Figure 11 PC1 vs PC2 Cluster Plot

The number of countries in each cluster was different.

# cluster	# count
0	38
1	64
2	79
3	25

Figure 12 Cluster size

	# 0	# 1	# 2	# 3
gdp_per_capit	1678.643036861872	55941.53456154049	9714.31147906034	4170.92773701419
inflation	17.75005617252213	2.527695601799422	7.326137404988494	8.80397064647852
unemploymer	5.303842105263158	5.196625	5.6753544303797465	16.149839999999998
life_expectanc	63.70992105263158	81.05018025914634	73.52156529793146	70.77306048780488
secondary_en	61.915265581946116	107.1892752772177	90.00015733780116	83.58513885909433
electricity_acc	47.29210526315789	100.0	97.05949367088607	87.772
drinking_wate	67.73092315476119	99.31709379343307	95.00958767545015	91.36953908273223
sanitation_acc	39.053369390048424	98.19323930723962	88.2214468411824	80.01586295733873
government_e	-0.9096347605263148	1.1498739734375	-0.1417910443037971	-0.7073830639999992
political_stabi	-1.0908268184210512	0.7420375656250001	0.17721075696202557	-1.005587259999999
rule_of_law	-0.9992607131578941	1.0652956890625	-0.2339044911392399	-0.8488423079999983
forest_area	28.355545945142342	29.940329101176808	42.39422413864948	13.846521682326973
pm25_exposu	39.139985580310785	16.129172805675754	19.385393616034918	31.443323583052557
renewable_en	70.57894736842105	18.734375	21.308860759493673	24.88

Cluster 1 had the strongest overall performance. Countries in this group had higher average GDP per capita, life expectancy, secondary enrolment, infrastructure access, and governance scores.

Cluster 0 had lower average infrastructure access, but it had higher renewable energy values. Cluster 3 had higher average unemployment and weaker governance. Cluster 2 showed more middle-level performance across many indicators.

This helped me to know that countries are not only different because of their final index score. They can also have different strengths and weaknesses across areas such as economy, health, education, infrastructure, governance, and environment.

Normalisation

I worked on normalisation because the indicators had different units and scales. If I combined these indicators directly, some indicators with larger numbers could have too much influence on the final index. So I changed all indicators into the same scale from 0 to 1.

Positive and Negative Indicators

I separated the indicators into positive and negative indicators.

Positive indicators are indicators where a higher value means better performance. These included GDP per capita, life expectancy, secondary enrolment, electricity access, drinking water access, sanitation access, government effectiveness, political stability, rule of law, forest area, and renewable energy.

Negative indicators are indicators where a lower value means better performance. These included inflation, unemployment rate, and PM2.5 exposure.

A screenshot of a code editor with a dark background. The title of the editor is "3. Define positive and negative indicators". There are two code blocks. The first block defines a list of positive indicators: "gdp_per_capita", "life_expectancy", "secondary_enrollment", "electricity_access", "drinking_water_access", "sanitation_access", "government_effectiveness", "political_stability", "rule_of_law", "forest_area", and "renewable_energy". The second block defines a list of negative indicators: "inflation", "unemployment_rate", and "pm25_exposure". Both blocks are followed by a green checkmark and "0.0s".

```
3. Define positive and negative indicators

positive_indicators = [
    "gdp_per_capita",
    "life_expectancy",
    "secondary_enrollment",
    "electricity_access",
    "drinking_water_access",
    "sanitation_access",
    "government_effectiveness",
    "political_stability",
    "rule_of_law",
    "forest_area",
    "renewable_energy"
]

negative_indicators = [
    "inflation",
    "unemployment_rate",
    "pm25_exposure"
]
```

Figure 13 Separated Positive and Negative indicators

Min-Max Normalisation

I used min-max normalisation for all selected indicators.

For positive indicators, I used this formula:

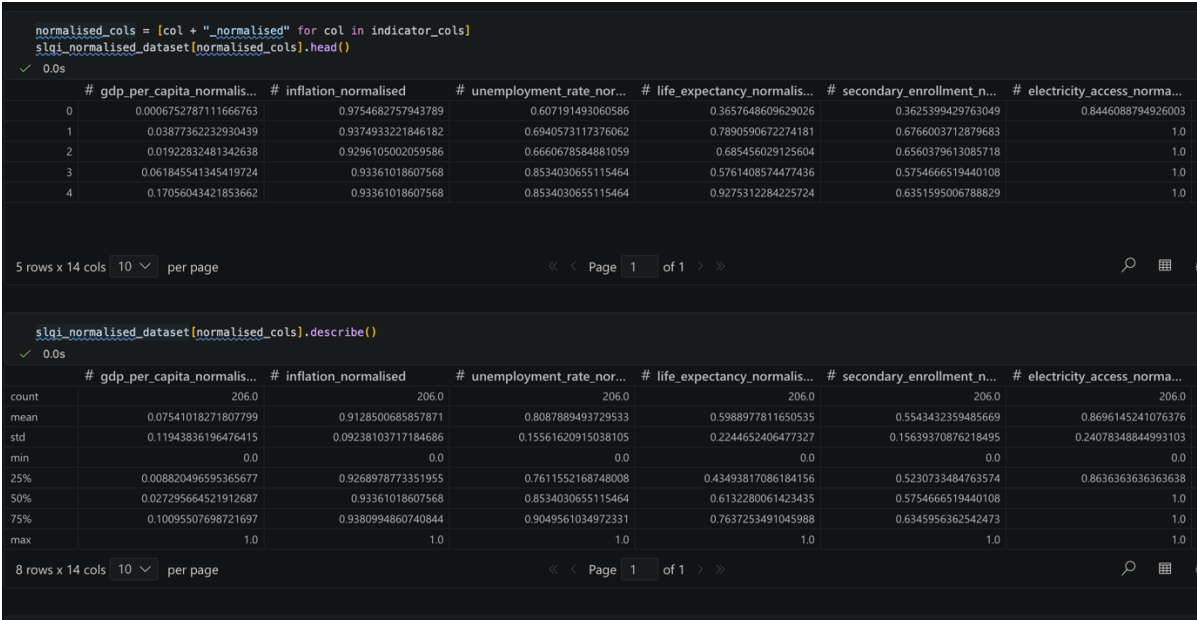
- $\text{normalised value} = (\text{value} - \text{minimum value}) / (\text{maximum value} - \text{minimum value})$

For negative indicators, I reversed the formula as

- $\text{normalised value} = (\text{maximum value} - \text{value}) / (\text{maximum value} - \text{minimum value})$

After this step, all indicators had the same meaning. A value closer to 1 meant better performance, and a value closer to 0 meant weaker performance.

After normalisation, I checked the dataset again using descriptive statistics. All values were between 0 and 1, so this showed that the normalisation worked correctly.



Weighting and Aggregation

After normalisation, I worked on calculating the dimension scores. I calculated each dimension score by taking the average of the normalised indicators in that dimension.

This step gave each country five dimension scores:

economic_score	GDP per capita, inflation, and unemployment rate.
health_education_score	life expectancy and secondary enrolment.
infrastructure_score	electricity access, drinking water access, and sanitation access.
governance_score	government effectiveness, political stability, and rule of law
environment score	forest area, PM2.5 exposure, and renewable energy.

Equal Weighting Method

I used equal weighting to calculate the final SLQI score. Each dimension was given the same weight of 20%.

Dimension	weight
Economic	20%
Health and Education	20%
Infrastructure	20%
Governance	20%
Environment	20%

I used equal weighting because there was no strong evidence to say which one dimension should be more important than another. It also made the final index easier to understand. So a higher SLQI score means the country had stronger overall sustainable living quality based on the selected indicators.

Final SLQI Ranking

After calculating the final SLQI score, I ranked the countries from highest to lowest. The top countries usually had strong scores in health and education, infrastructure, and governance. Monaco ranked first, followed by Finland, Sweden, Liechtenstein, and Norway...

	Country	Country Code	# slqi_rank	# slqi_score
107	Liechtenstein	LIE	1	0.8437076622032934
123	Monaco	MCO	2	0.8391431704799408
63	Finland	FIN	3	0.8367616576858559
177	Sweden	SWE	4	0.8276054566057456
139	Norway	NOR	5	0.8160416164062445
83	Iceland	ISL	6	0.7992559360301501
109	Luxembourg	LUX	7	0.7911412723892308
178	Switzerland	CHE	8	0.7873041414753735
50	Denmark	DNK	9	0.7842923893984318
92	Japan	JPN	10	0.7832035623499791

Figure 14 Top 10 countries by final SLQI score

The lowest ranked countries usually had weaker scores in health and education, infrastructure, or governance. Chad, Niger, South Sudan, Burkina Faso, and Central African Republic were among the lowest ranked countries.

	Country	# slqi_score	# slqi_rank	# economic_score
175	Sudan	0.39010904256025386	197	0.3791459554396639
118	Mauritania	0.38786965157594233	198	0.549816602664187
0	Afghanistan	0.3781339484596874	199	0.5277783491887105
42	Congo, Dem. Rep.	0.37337994826660303	200	0.6035914219335471
36	Central African Republic	0.3578842846400534	201	0.5880979660962115
29	Burkina Faso	0.34320639118188584	202	0.611900737773731
167	Somalia, Fed. Rep.	0.33801204930128903	203	0.4632541587228836
136	Niger	0.30377447953692394	204	0.6342102952871226
37	Chad	0.2879593075717271	205	0.6281807558878708
169	South Sudan	0.27391877526576697	206	0.40857782981690977

Figure 15 bottom 10 countries by final SLQI score

The bottom countries did not always have the lowest score in every dimension. For example, some countries may have better environmental scores, but weaker infrastructure or governance scores. This shows why the dimension scores help explain the final ranking in more detail.

Sensitivity Analysis

Sensitivity analysis to test whether the rankings changed when I used different weighting methods. I tested three different weighting scenarios:

- Social focus
The social focus scenario gave more weight to health and education and infrastructure.
- Environment focus
The environment focus scenario gave more weight to the environment dimension.
- Governance focus
The governance focus scenario gave more weight to the governance dimension.

Scenario	Average Rank Change	Maximum Rank Change
Social focus	7.61	34
Environment focus	16.35	85
Governance focus	7.5	34

The results showed that the SLQI ranking was stable in the social focus and governance focus scenarios. The average rank change was around 7.5 places, and the maximum rank change was 34 places.

The environment focus scenario changed the rankings more. The average rank change was 16.35 places, and the maximum rank change was 85 places. This means the index was more sensitive when the environment dimension was given more importance. Maybe because environmental scores did not strongly match the other dimensions.

So, the sensitivity analysis showed that the SLQI ranking was mostly stable. Only the environmental performance had a higher weight, some countries changed position.

Link to Other Indices

I compared the SLQI index with the Human Development Index (HDI) to check if my index had a reasonable relationship with an existing development index. This comparison was not used to prove that the SLQI is perfect. It was only used to see if countries with higher HDI values also usually had higher SLQI scores.

To do this, I merged the final SLQI dataset with the HDI dataset. Before merging, I cleaned and mapped some country names because different datasets used different country name formats.

Country Matching

After cleaning and mapping the country names, 193 countries were matched between the SLQI dataset and the HDI dataset. Some countries could not be matched. This was because of different country names or missing HDI values.

Correlation Between SLQI and HDI

I calculated the Pearson correlation between the SLQI score and the 2023 HDI value. The result showed a strong positive relationship.

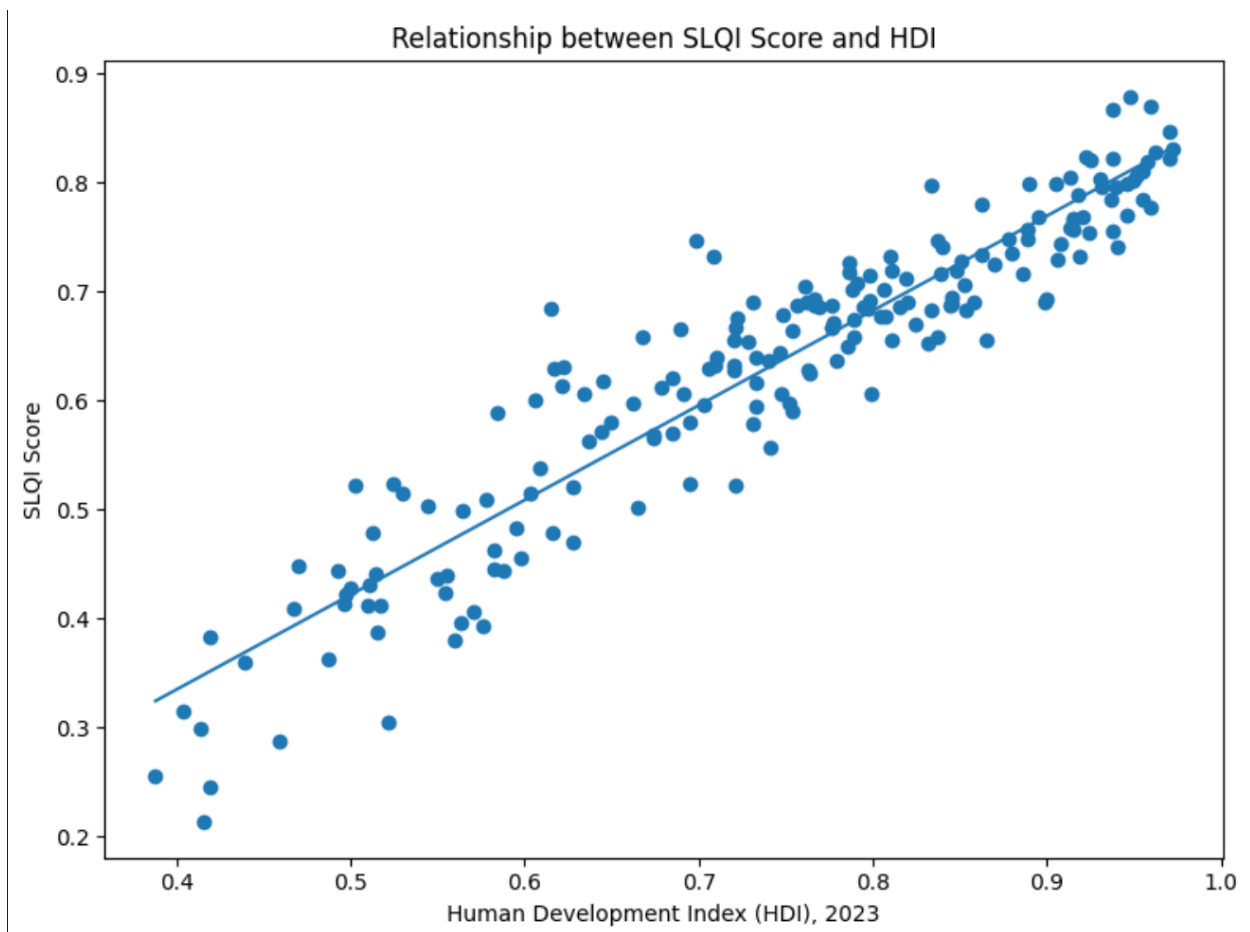


Figure 16 Relationship between SLQI Score and HDI

	Result
Matched countries	193
Person correlation coefficient	0.937159
P- value	2.79e-89
R square	0.878267

The Pearson correlation coefficient was 0.937. This means there was a very strong positive relationship between SLQI and HDI. The p-value was very small, so the relationship was statistically significant. The R-squared value was 0.878. This means that HDI explained a large part of the variation in SLQI scores in the linear regression model.

However, SLQI and HDI are not exactly the same. HDI focuses on human development. My SLQI also includes other areas. Because of this, the SLQI gives a broader view of sustainable living quality, but it is still a simplified index.

Visualisation of SLQI Results

I used visualisation to make the SLQI results easier to understand. Ranking tables are useful, but they can be hard to read when there are many countries.

For this part, I used the final SLQI index score dataset. The main visualisations were:

- Top 20 countries by SLQI score
- Bottom 20 countries by SLQI score
- Dimension scores of top 20 countries
- World map of SLQI scores
- World map of SLQI categories

Top 20 Countries by SLQI Score

I created a bar chart to show the top 20 countries with the highest SLQI scores.

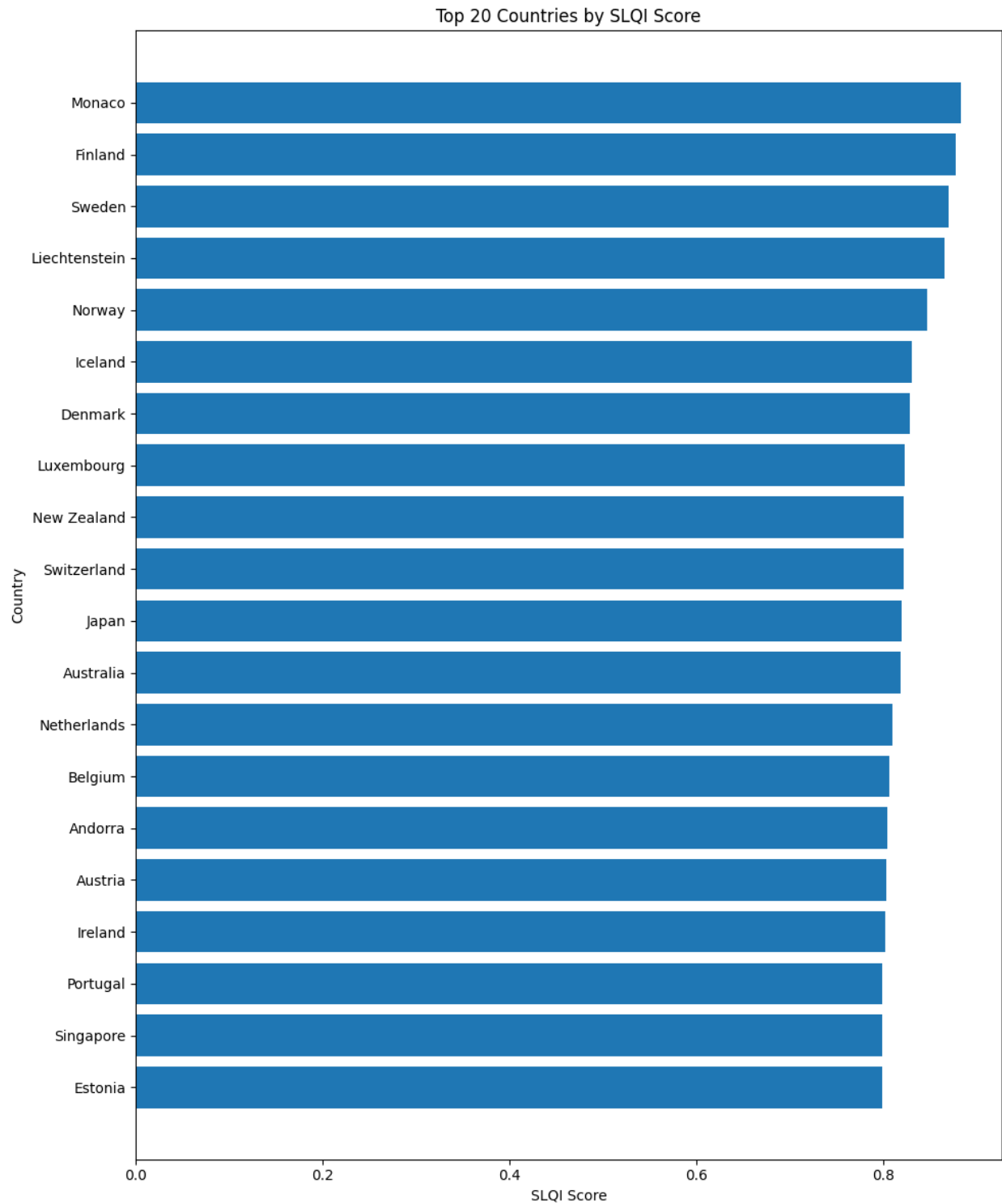


Figure 17 Top 20 Countries by SLQI Score

The top countries usually had strong scores in infrastructure, health and education, and governance.

Some of these countries did not have the highest environmental scores. However, their overall performance across the five dimensions was still strong.

Bottom 20 Countries by SLQI Score

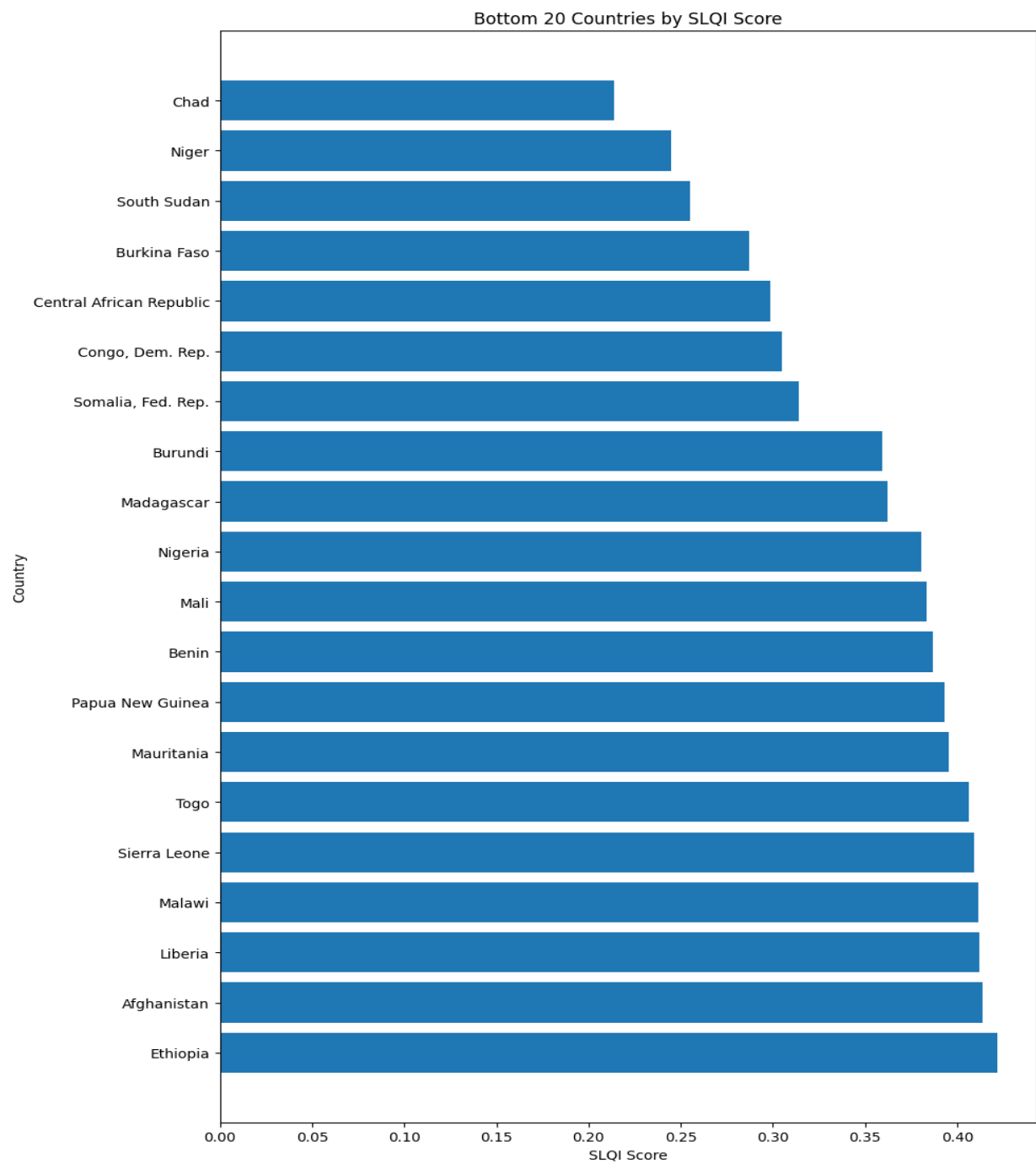


Figure 18 Bottom 20 Countries by SLQI Score

I also created a bar chart to show the bottom 20 countries with the lowest SLQI scores.

The lowest ranked countries usually had weaker scores in several areas. These areas were infrastructure, health and education, and governance.

This showed that a low SLQI score was usually not caused by only one weak indicator. It was often linked to several challenges at the same time.

Dimension Scores of Top 20 Countries

I used a grouped bar chart to compare the five dimension scores for the top 20 countries.

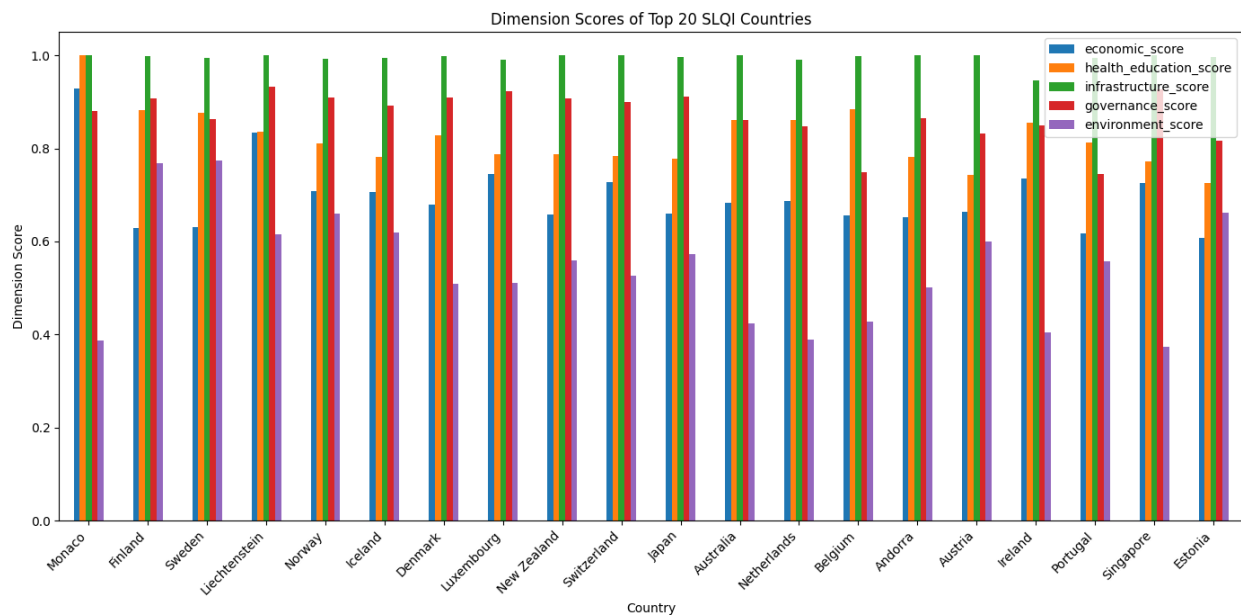


Figure 19 Dimension Scores of Top 20 SLQI Countries

This chart compared the strengths and weaknesses of the top countries. Many high-ranking countries had very high infrastructure scores and strong governance scores. However, the environmental scores were more different between countries.

This matched the earlier correlation result, where the environment score had a weak relationship with the final SLQI score.

World Map of SLQI Scores

I created a world map to show SLQI scores across the world.

World Map of Sustainable Living Quality Index (SLQI)

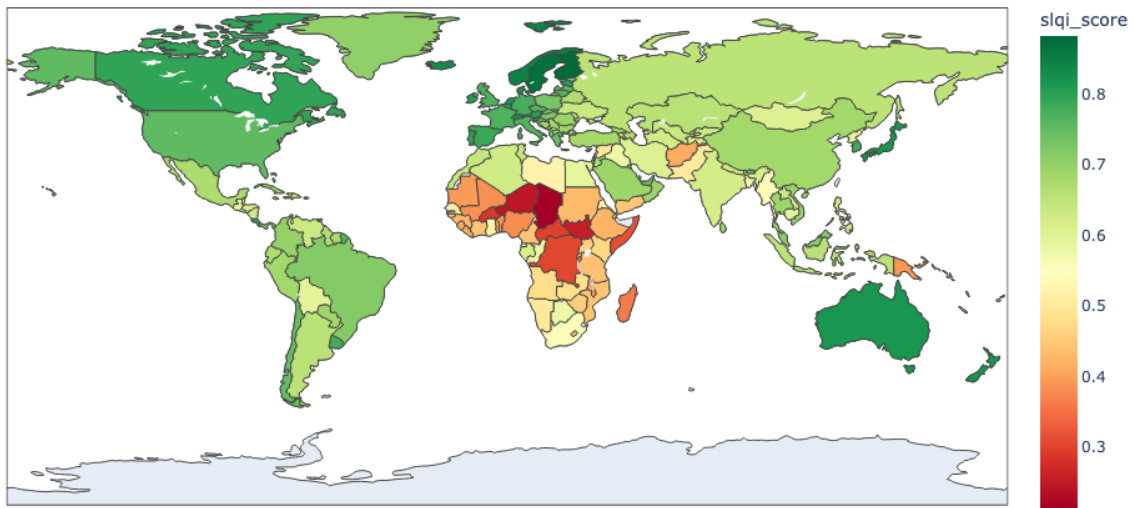


Figure 20 World Map of Sustainable Living Quality Index

The map made it easier to see regional patterns.

Countries with higher scores appeared in different parts of the world. However, many high-scoring countries were advanced economies with strong infrastructure and governance systems.

World Map of SLQI Categories

For this, I grouped countries into SLQI categories based on their final score.

The categories were:

- Very low
- Low
- Medium
- High
- Very High

This category map gave a simpler view of the results. It was useful because it showed the general differences between countries more clearly.

World Map of SLQI Categories

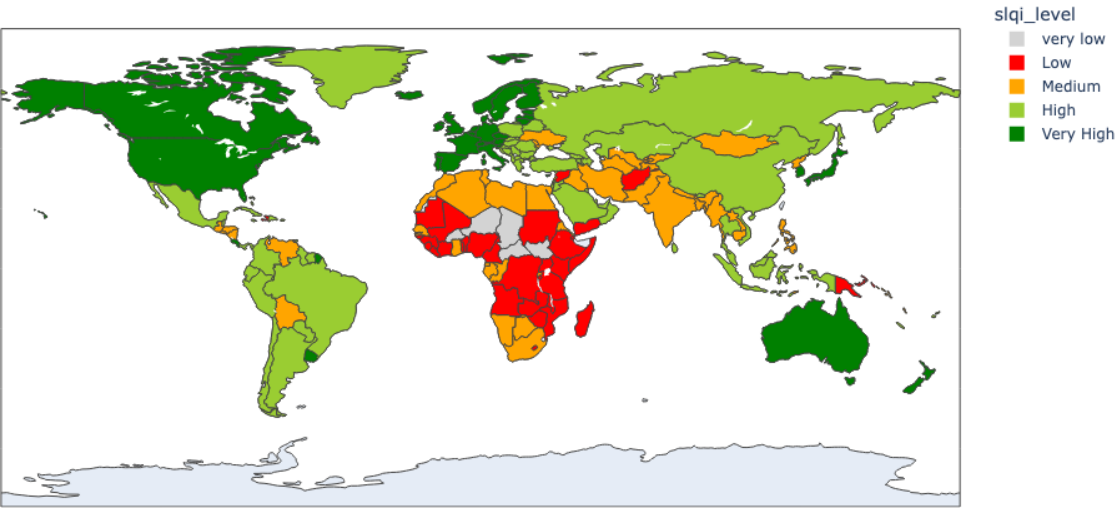


Figure 21 World Map of SLQI Categories

Countries rank result:

1	Country	Country Cod	slqi_rank	slqi_score	economic_score	health_education_score	infrastructure_score	governance_score	environment_score
2	Liechtenstein	LIE	1	0.843707662	0.834927604	0.835667636	0.999814076	0.932930338	0.615198658
3	Monaco	MCO	2	0.83914317	0.929004417	1	1	0.879638631	0.387072804
4	Finland	FIN	3	0.836761658	0.628199644	0.882677214	0.997947178	0.907696374	0.767287878
5	Sweden	SWE	4	0.827605457	0.630971918	0.876168525	0.994086888	0.863664893	0.773135059
6	Norway	NOR	5	0.816041616	0.707387577	0.811396665	0.992737987	0.909410052	0.659275801
7	Iceland	ISL	6	0.799255936	0.706495196	0.782228619	0.99545602	0.892554666	0.619545179
8	Luxembourg	LUX	7	0.791141272	0.744846907	0.787200247	0.991080005	0.922393021	0.510186181
9	Switzerland	CHE	8	0.787304141	0.727002503	0.783191954	0.999627655	0.900504754	0.526193841
10	Denmark	DNK	9	0.784292389	0.678563978	0.827357828	0.998502291	0.908626603	0.508411247
11	Japan	JPN	10	0.783203562	0.659565925	0.777719456	0.995555943	0.910983679	0.572192808
12	New Zealand	NZL	11	0.782157608	0.657185871	0.786812807	1	0.906661194	0.560128167
13	Austria	AUT	12	0.767399693	0.663065204	0.742741319	0.999896249	0.831715439	0.599580252
14	Australia	AUS	13	0.765263698	0.682224804	0.86018144	0.999842637	0.860622011	0.423447599
15	Estonia	EST	14	0.761522632	0.607961748	0.7258741	0.996361424	0.816344805	0.661071082
16	Andorra	AND	15	0.759923169	0.652524562	0.781345365	1	0.864926653	0.500819268
17	Singapore	SGP	16	0.759456312	0.725175081	0.771871353	1	0.926500858	0.373734269
18	Ireland	IRL	17	0.758310671	0.736213751	0.855256896	0.946755308	0.848964762	0.404362636
19	Canada	CAN	18	0.755940743	0.64819697	0.756427904	0.985888292	0.841144431	0.548046118
20	Netherlands	NLD	19	0.75533973	0.687680328	0.861136275	0.991286362	0.846912802	0.389682881
21	Slovenia	SVN	20	0.753230728	0.651186801	0.769861341	0.991515389	0.762394707	0.591195403
22	Costa Rica	CRI	21	0.745908711	0.605146434	0.795323607	0.994539693	0.708761325	0.625772494
23	Portugal	PRT	22	0.745612244	0.617695476	0.812372684	0.9950407	0.744993883	0.557958476
24	Belgium	BEL	23	0.742707826	0.656024807	0.88392369	0.99808952	0.748154039	0.427347075
25	Korea, Rep.	KOR	24	0.739296935	0.661766746	0.754311018	0.998993606	0.796815797	0.484597505
26	Germany	DEU	25	0.73752365	0.678920389	0.72265095	0.997125437	0.801738991	0.487182484
27	Bhutan	BTN	26	0.735464832	0.618789009	0.559168949	0.948722731	0.758180806	0.792462664
28	Macao SAR, CHN	MAC	27	0.731626077	0.710603393	0.744511574	1	0.780018656	0.422996761
29	Lithuania	LTU	28	0.731504822	0.616852947	0.681070428	0.986176768	0.818799048	0.554624919
30	Hong Kong SAR, CHN	HKG	29	0.731498869	0.681305635	0.817120384	0.980791704	0.791970759	0.386305865
31	Uruguay	URY	30	0.73114107	0.591458431	0.744095089	0.994938647	0.773170492	0.55204269
32	Czechia	CZE	31	0.72634468	0.658266599	0.701399695	0.989981697	0.804810013	0.477265396
33	Spain	ESP	32	0.724363838	0.576845976	0.830159356	0.999305283	0.704614131	0.510894443
34	Latvia	LVA	33	0.724291809	0.608669319	0.65742781	0.965879066	0.73761675	0.651866098
35	Bermuda	BMU	34	0.722966561	0.760884041	0.66601593	0.999783036	0.796485178	0.391664618
36	San Marino	SMR	35	0.718822702	0.667466949	0.67305077	0.99999998	0.812312625	0.441283186
37	United Kingdom	GBR	36	0.718055526	0.664871917	0.77023682	0.996663255	0.756884222	0.401621414
38	United States	USA	37	0.716841187	0.704848413	0.683871876	0.998479208	0.718548304	0.478458134
39	Brunei Darussalam	BRN	38	0.715531861	0.638943452	0.61253036	1	0.749594733	0.576590759
40	France	FRA	39	0.714721456	0.62907362	0.768393699	0.994959122	0.696703163	0.484477678
41	Guam	GUM	40	0.714330257	0.643267966	0.645322303	0.961776059	0.801204224	0.52008073
42	Cayman Islands	CYM	41	0.712714625	0.710253291	0.668803232	0.914165844	0.814705833	0.455644922
43	Italy	ITA	42	0.710979281	0.63260283	0.773618454	0.999142045	0.680096674	0.4694364
44	Samoa	WSM	43	0.707663699	0.606250792	0.557506096	0.986315664	0.759043948	0.629201995
45	Palau	PLW	44	0.704053123	0.614770314	0.529051447	0.993949846	0.736597651	0.645896356
46	Virgin Islands	VIR	45	0.703816615	0.586548766	0.697872861	0.994475037	0.696658612	0.543527801
47	Cyprus	CYP	46	0.703199088	0.644899898	0.759721092	0.996612386	0.697342516	0.417419552
48	Croatia	HRV	47	0.702178581	0.625313229	0.708813313	0.973436433	0.675834775	0.527495154
49	Malta	MLT	48	0.697600125	0.668550397	0.765568345	1	0.7140744	0.339807481
50	St. Kitts and Nevis	KNA	49	0.695949904	0.622538986	0.709150134	0.958891089	0.716696954	0.472472358
51	American Samoa	ASM	50	0.693364617	0.616286264	0.575803755	0.859142726	0.787738278	0.627852062
52	Seychelles	SYC	51	0.692144423	0.62013061	0.55729334	0.987106267	0.712699595	0.583492305
53	Slovak Republic	SVK	52	0.692035811	0.624973229	0.661928977	0.989754846	0.690470867	0.493051136
54	Aruba	ABW	53	0.691985012	0.641167522	0.733044556	0.980639456	0.795922652	0.309150876
55	French Polynesia	PYF	54	0.691352498	0.559369264	0.751561711	0.990740566	0.708435483	0.446655464
56	Puerto Rico	PRI	55	0.690532743	0.63664677	0.67961634	1	0.605871637	0.530528971
57	Chile	CHL	56	0.687261024	0.578998449	0.748821793	0.995238073	0.684240013	0.429006794
58	Montenegro	MNE	57	0.685279385	0.533039877	0.65657798	0.98827131	0.615472804	0.633034955
59	Poland	POL	58	0.685035157	0.646538233	0.709862357	0.944938272	0.68247395	0.441362971
60	New Caledonia	NCL	59	0.684813675	0.573222078	0.668845623	0.997609159	0.718969088	0.465422428
61	United Arab Emirates	ARE	60	0.682097761	0.684995032	0.764448619	0.996604572	0.74238358	0.222057
62	Greenland	GRL	61	0.681254586	0.663175357	0.533141879	0.973938199	0.869085343	0.366932154
63	Greece	GRC	62	0.6801952	0.577827534	0.755011836	0.996308524	0.596771268	0.475056839
64	Malaysia	MYS	63	0.676816509	0.623989152	0.59599435	0.974537199	0.673504988	0.516056855
65	Hungary	HUN	64	0.672803872	0.628213744	0.681705746	0.992464968	0.634511397	0.427123504
66	Panama	PAN	65	0.672789018	0.589591243	0.682247167	0.924176244	0.566141045	0.601789391
67	Antigua and Barbuda	ATG	66	0.672623475	0.618261923	0.702980067	0.992950872	0.710406179	0.338518334
68	Albania	ALB	67	0.672116014	0.556774752	0.732829719	0.972102282	0.563813459	0.535059856
69	Micronesia, Federated States of	FSM	68	0.671867602	0.596952841	0.487375533	0.912731822	0.726766967	0.635510846
70	Dominica	DMA	69	0.671456022	0.608256291	0.581136726	0.958366308	0.685648927	0.523871859

71	Israel	ISR	70	0.670483896	0.673630154	0.754924805	0.999702809	0.60346945	0.320692261
72	Fiji	FJI	71	0.670065211	0.600093071	0.506542958	0.947582004	0.671636638	0.624471381
73	Brazil	BR	72	0.669958234	0.590000711	0.660369502	0.971039521	0.456193021	0.672188416
74	Barbados	BRB	73	0.669094306	0.61640762	0.665948319	0.988567181	0.748729291	0.325819118
75	Grenada	GRD	74	0.668588062	0.611891023	0.63907959	0.93777545	0.68462471	0.469569537
76	St. Vincent and the Grenadines	VCT	75	0.662900546	0.483381541	0.654315891	0.958891089	0.687727448	0.530186759
77	Mauritius	MUS	76	0.661576101	0.605143943	0.582077562	0.989853614	0.71666429	0.414141096
78	Bahamas, The	BHS	77	0.657463581	0.607729942	0.560322424	0.958891089	0.701820582	0.45855387
79	Qatar	QAT	78	0.655042235	0.735766018	0.747646659	0.999706892	0.752789402	0.039302202
80	Romania	ROU	79	0.654669632	0.612904085	0.594674491	0.964828206	0.620992938	0.479948443
81	Viet Nam	VNM	80	0.650356023	0.634806629	0.60559699	0.975945246	0.516969686	0.518461563
82	Paraguay	PRY	81	0.64963547	0.597022579	0.547753451	0.982124368	0.469956894	0.651320057
83	Tonga	TON	82	0.646218948	0.632943561	0.576274687	0.971524997	0.696693174	0.353658323
84	Bulgaria	BGR	83	0.645826507	0.62631815	0.626093	0.944003463	0.551848737	0.480869184
85	Sri Lanka	LKA	84	0.64420638	0.614919394	0.634083762	0.932003229	0.480373271	0.559652245
86	Colombia	COL	85	0.642205729	0.556795851	0.681746102	0.96766248	0.421050538	0.583773673
87	Tuvalu	TUV	86	0.641549116	0.60276598	0.479386573	0.95508261	0.705783478	0.464726937
88	Suriname	SUR	87	0.640992891	0.561172134	0.534494308	0.951023099	0.534531755	0.623743159
89	Thailand	THA	88	0.640548733	0.64901935	0.631367526	0.9975848	0.497624786	0.427147204
90	Cuba	CUB	89	0.640367103	0.639713646	0.655591312	0.955399737	0.503120134	0.448010688
91	Georgia	GEO	90	0.637575111	0.54750508	0.628671161	0.951301715	0.546967415	0.513430183
92	St. Lucia	LCA	91	0.637356929	0.568094504	0.558073321	0.964069011	0.687844229	0.408703579
93	Peru	PER	92	0.636765424	0.606708658	0.703819919	0.885067844	0.442964937	0.545265763
94	Serbia	SRB	93	0.633370995	0.589227526	0.613426926	0.968500443	0.527080981	0.468619097
95	Kuwait	KWT	94	0.63325484	0.662974057	0.757658124	1	0.613565252	0.132076769
96	Guyana	GUY	95	0.632987326	0.565535609	0.529897428	0.941231922	0.53114361	0.59712806
97	Trinidad and Tobago	TTO	96	0.632844329	0.638910051	0.595347581	0.969885249	0.554928514	0.40515025
98	Oman	OMN	97	0.632548341	0.641722423	0.697571713	0.959488082	0.67437884	0.189580646
99	Dominican Republic	DOM	98	0.631669715	0.606565638	0.537054232	0.948672783	0.58475383	0.481302092
100	Belize	BLZ	99	0.63049483	0.570181137	0.567544046	0.913513038	0.567584666	0.533651262
101	Jamaica	JAM	100	0.630449604	0.620792166	0.527628475	0.896268776	0.591579325	0.515979276
102	China	CHN	101	0.629889195	0.620778307	0.654659445	0.969759823	0.557780239	0.346468163
103	Marshall Islands	MHL	102	0.62959903	0.604366221	0.460708719	0.856690337	0.690584501	0.535645372
104	North Macedonia	MKD	103	0.628272285	0.537308505	0.624348471	0.986878331	0.540027298	0.452798819
105	Maldives	MDV	104	0.626295203	0.622183452	0.663085051	0.997637601	0.530942483	0.317627427

106	Saudi Arabia	SAU	105	0.624550936	0.654293808	0.727805596	0.986132344	0.619726074	0.134796857
107	Ecuador	ECU	106	0.623445154	0.622405437	0.648276516	0.921619838	0.400179752	0.524744225
108	Bosnia and Herzegovina	BIH	107	0.62121026	0.555178455	0.639641061	0.953660577	0.436923238	0.520647969
109	Indonesia	IDN	108	0.620591279	0.620705189	0.568963339	0.89705707	0.498508416	0.517722382
110	Bahrain	BHR	109	0.619847604	0.672571249	0.727936024	0.999584874	0.58516018	0.113985692
111	Mexico	MEX	110	0.618004719	0.633800714	0.645071692	0.970504241	0.385697844	0.454949102
112	Lao PDR	LAO	111	0.617980215	0.607927529	0.385221896	0.924328191	0.488988546	0.683434914
113	Belarus	BLR	112	0.612965784	0.617944139	0.604219599	0.998503372	0.369604697	0.474557111
114	Moldova	MDA	113	0.61181773	0.638291257	0.571288727	0.91343267	0.528225355	0.407850643
115	Solomon Islands	SLB	114	0.611145043	0.632102232	0.539741988	0.543965124	0.552957175	0.786958696
116	Cabo Verde	CPV	115	0.607361214	0.539754966	0.638674734	0.892319729	0.671942641	0.294114001
117	Russian Federation	RUS	116	0.605407621	0.631680445	0.582276269	0.95499674	0.363430807	0.494653843
118	Turkiye	TUR	117	0.603078528	0.499434421	0.728272527	0.97651471	0.401715092	0.409455891
119	Azerbaijan	AZE	118	0.601983475	0.600725106	0.595710008	0.98422002	0.511557837	0.317704404
120	Kazakhstan	KAZ	119	0.598464308	0.607253702	0.614578996	0.981130397	0.507345701	0.282012745
121	Gabon	GAB	120	0.598275263	0.463819463	0.50548488	0.761725468	0.393297181	0.867049324
122	Vanuatu	VUT	121	0.597538962	0.589735026	0.690153107	0.603890549	0.594163043	0.509753083
123	Philippines	PHL	122	0.592713541	0.628749766	0.504276052	0.923720748	0.454199882	0.45262126
124	Nepal	NPL	123	0.592147806	0.543552881	0.531513076	0.893935942	0.426137285	0.565599847
125	Jordan	JOR	124	0.590917029	0.491903811	0.651170188	0.982968696	0.550740002	0.27780245
126	Honduras	HND	125	0.59069242	0.599947841	0.444867493	0.916521221	0.365370817	0.62675473
127	Cambodia	KHM	126	0.590610981	0.649444337	0.432553275	0.831535239	0.452807349	0.586714703
128	Timor-Leste	TLS	127	0.589699891	0.633885586	0.470448516	0.833697753	0.472053569	0.538414034
129	Nicaragua	NIC	128	0.587680127	0.598574643	0.536445815	0.917664874	0.336294023	0.549421281
130	Morocco	MAR	129	0.586614435	0.565491284	0.606429681	0.91419136	0.498486555	0.348473292
131	Armenia	ARM	130	0.584579482	0.539727435	0.653885709	0.976088565	0.454074647	0.299121052
132	El Salvador	SLV	131	0.58372851	0.623755472	0.466660017	0.906111309	0.478701096	0.443414656
133	India	IND	132	0.582399275	0.605720496	0.517036924	0.914283566	0.5149488	0.36006591
134	Guatemala	GTM	133	0.580053278	0.627780126	0.433143981	0.857473645	0.388095539	0.593773097
135	Uzbekistan	UZB	134	0.578049919	0.596980256	0.58199011	0.973121013	0.483643585	0.254514629
136	Ukraine	UKR	135	0.577256445	0.551942947	0.581572431	0.957748718	0.413313155	0.381704972
137	Nauru	NRU	136	0.577240546	0.61118802	0.403995409	0.882179201	0.65379568	0.335052239
138	Kyrgyz Republic	KGZ	137	0.576942479	0.611925386	0.562262472	0.968362889	0.369046475	0.373115174
139	Kiribati	KIR	138	0.574009784	0.599018244	0.467825715	0.673202856	0.672926252	0.457075853
140	Venezuela, Republic of	VEN	139	0.573563903	0.599032043	0.587329022	0.959474881	0.130702385	0.591281183

Conclusion

In this report, I built the Sustainable Living Quality Index, or SLQI. The index used 14 indicators and grouped them into five dimensions: economic conditions, health and education, infrastructure, governance, and environment.

I started by cleaning and merging the datasets. I removed countries with too many missing values and filled the remaining missing values using median imputation. The final dataset included 206 countries.

I then used correlation analysis, PCA, and cluster analysis to understand the data. The results showed that sustainable living quality is made up of different areas, not just one factor.

After that, I normalised the indicators to a 0 to 1 scale. I calculated the dimension scores and used equal weighting to produce the final SLQI score.

The final ranking showed that countries such as Monaco, Finland, Sweden, Liechtenstein, and Norway had the highest SLQI scores. These countries usually performed well in infrastructure, health and education, and governance.

I also compared the SLQI with HDI. The strong correlation showed that the SLQI had a reasonable link with an existing development index. However, SLQI is broader because it also includes governance, infrastructure, environment, and economic indicators.

Overall, the SLQI gives a simple and clear way to compare countries based on sustainable living quality. It is not perfect, but it provides a useful overview based on the selected indicators.

GitHub Link:

Read more detail: https://github.com/EricHanJf/DVA_CA1_SD4a_2026

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